

Tamilselvan P

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Professional Summary

Entry-level IT professional with hands-on experience in SQL, MySQL, Python, database-driven applications, data validation, ETL, and troubleshooting. Strong foundation in Database Management Systems, SQL querying, data integrity, and problem-solving, with experience processing and validating large-scale datasets using Python and MySQL. Eager to build expertise in Oracle Database Administration and contribute to database monitoring, incident support, backup and recovery, user access management, and routine DBA operations in an L1 support environment.

Education

Bachelor of Engineering (B.E.) – Electronics and Communication Engineering

Sept 2021 – May 2025

Shree Venkateshwara Hi-tech Engineering College, Gobi

- GPA: 7.8/10.0
- **Coursework:** Database Management Systems, SQL, Python Programming, Data Structures

Technical Skills

- **Programming:** Python, SQL
- **Databases:** MySQL, Database Management Systems (DBMS), Joins, Subqueries, Aggregations, Window Functions
- **Database Concepts:** Data Validation, Data Integrity, Database Connectivity, Database-Driven Applications
- **Data Processing & Automation:** ETL, Data Cleaning, Data Validation, Web Scraping
- **Software Development:** Object-Oriented Programming (OOP), Data Structures & Algorithms, Exception Handling, File Handling, Debugging
- **Tools:** Git, GitHub, VS Code, PyCharm

Experience

Junior Data Analyst | Pathfinder Business Analysis (P) Ltd., Coimbatore

Sep 2025 – Dec 2025

- Developed and maintained Python-based ETL workflows to extract, transform, validate, and process large-scale web datasets.
- Enhanced existing Python applications by improving data collection logic, debugging application issues, and optimizing processing performance across multiple websites.
- Developed reusable Python components and integrated them with the organization's internal data collection platform.
- Processed, cleaned, and validated data collected from 20,000+ websites using Python and SQL, ensuring data accuracy and consistency.
- Loaded structured datasets into MySQL and performed SQL-based validation to maintain data integrity and consistency.
- Collaborated with team members using Git-based version control during feature development and application enhancements.

Projects

Secure Database Connection Framework | Python, MySQL, REST APIs

- Developed a secure database connection framework to connect Python applications with MySQL while preventing database credentials from being exposed in source code.
- Implemented encrypted credential management and runtime decryption to securely retrieve database credentials when required.
- Integrated secure credential handling with Python-MySQL connectivity to improve application security and reliability.

URL Shortener Application | Python, Flask, MySQL, REST APIs

- Developed a web-based URL Shortener application using Python and Flask to convert long URLs into short, easy-to-use links.
- Designed and implemented REST API endpoints for URL creation, retrieval, and redirection, with MySQL for persistent data storage.
- Implemented URL redirection logic by retrieving the original URL from MySQL when a shortened URL is accessed.

Certifications

- HackerRank SQL (Advanced)
- HackerRank SQL (Intermediate)
- Python Programming